Loyal Circle

Time limit: 2 seconds

A university has launched its own messaging app, Baithak, for its N students. The students are connected by M friendships; friendship is mutual, so if student U is a friend of student V then V is also a friend of U.

The product team has noticed a simple rule about how students behave. A student keeps using the app only while at least K of their friends are still using it. At the start every student is active. As soon as an active student has fewer than K active friends, that student leaves the app and never comes back. One departure can leave other students with too few active friends, so they leave as well, and the chain reaction continues until every remaining student has at least K active friends, or until nobody is left.

The students who remain at the end are called the **loyal circle** for the threshold K. It can be shown that the loyal circle does not depend on the order in which students leave.

The team wants to study several thresholds. For each of Q given values of K, find the size of the loyal circle. Every value of K is studied separately, starting again from the complete network with all N students active.

Input

The first line contains two integers N ($1 \le N \le 100\,000$) and M ($0 \le M \le 200\,000$), the number of students and the number of friendships. Students are numbered from 1 to N.

Each of the next M lines contains two integers U and V ($1 \le U, V \le N$, $U \ne V$), meaning that students U and V are friends. No friendship is listed more than once.

The next line contains an integer Q ($1 \le Q \le 100\,000$), the number of thresholds to study.

Each of the next Q lines contains one integer K ($0 \le K \le 1\,000\,000\,000$).

Output

For each threshold K, in the order given, print one line with a single integer: the number of students in the loyal circle.

Sample Input	Sample Output
8 10 1 2 1 3 1 4 2 3 2 4 3 4 5 3	5 4 7 0 8

5 4 6 5 6 7 5 2 3 1 4 0	

Explanation

For K = 2, student 8 (no friends) and student 7 (one friend) leave at once. Student 6 is then left with a single active friend, student 5, and leaves too. Students 1, 2, 3, 4 and 5 each still have at least two active friends, so the loyal circle has 5 students.

For K = 3, the same students leave, and after that student 5 has only two active friends (3 and 4) even though student 5 started with three friends, so student 5 also leaves. Students 1, 2, 3 and 4 remain.

For K = 1 only student 8 leaves. For K = 4 everybody eventually leaves. For K = 0 nobody leaves.